

Name: KEY

Math 3108 – Linear Algebra
Section 101 – Brigham
Exam 2
October 18, 2024

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned off or completely silenced (i.e. not on vibrate) for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 4) Failure to follow directions specific to a problem will result in the loss of points.
- 5) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 6) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 7) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 8) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 9) If you need extra room, you may use the back of the previous page. However, you must indicate you are doing so by clearly writing "BPP" on the relevant problem.
- 10) This packet has six sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 11) Once this exam begins, you will have 50 minutes to complete your solutions.

Page	1	2	3	4	5	Sum
Points Earned						
Maximum Points	15	20	21	22	22	100

DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO

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Multiple Choice and True/False. Circle the correct answer.

1. (10 pts) Let A be a 5×5 matrix with a nontrivial null space. Label the following as True or False.

A is not invertible

- i) T There exists a $\mathbf{b} \in \mathbb{R}^5$ such that $A\mathbf{x} = \mathbf{b}$ has no solution.
ii) T $\det A^T = 0$.
iii) F The column space of A spans $\mathbb{R}^5$.
iv) F A has a pivot in every row.
v) T A is singular

2. (2 points) The set $\left\{ \begin{bmatrix} a \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ a \end{bmatrix} \right\}$ forms a basis for $\mathbb{R}^2$.

- i) Always true ii) Sometimes true iii) Never true

if $a = 0$

3. (3 points) Let A be an $m \times n$ matrix. Which of the following is a subspace of $\mathbb{R}^m$?

- i) Col A (the column space of A)
ii) Nul A (the null space of A)
iii) Both Col A and Nul A .
iv) Neither Col A nor Nul A .

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Open Answer. Show ALL your work!

4. (8 points) Use the method of your choice to calculate A^{-1} given $A = \begin{bmatrix} 3 & 5 \\ -6 & -9 \end{bmatrix}$. Then, use

the inverse you found to solve $A\mathbf{x} = \mathbf{b}$ given $\mathbf{b} = \begin{bmatrix} 1 \\ -6 \end{bmatrix}$. $\det A = -27 + 30 = 3$

$$A^{-1} = \frac{1}{3} \begin{bmatrix} -9 & -5 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} -3 & -5/3 \\ 2 & 1 \end{bmatrix}$$

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -3 & -5/3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -6 \end{bmatrix} = \begin{bmatrix} 7 \\ -4 \end{bmatrix}$$

5. (12 points) Find the LU -factorization of the matrix $A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \end{bmatrix}$.

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 10 & 4 & -9 \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \end{bmatrix} = U$$

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6. (12 points) Given the basis $B = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\}$ for P_2

a. Find the vector $\mathbf{p}(t) \in P_2$ such that $[\mathbf{p}(t)]_B = \begin{bmatrix} 5 \\ -4 \\ 1 \end{bmatrix}$

$$\mathbf{p}(t) = 5\vec{b}_1 - 4\vec{b}_2 + 1\vec{b}_3 = 5 + 5t - 4 - 4t^2 - 3t + 5t^2 = 1 + 2t + t^2$$

- b. Find $[\mathbf{p}(t)]_B$ if $\mathbf{p}(t) = -2$

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 1 & 6 & -3 & 0 \\ 0 & 1 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & -1 & -3 & -2 \\ 0 & 1 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & -1 & -3 & -2 \\ 0 & 0 & 2 & -2 \end{bmatrix}$$

$$\begin{aligned} c_3 &= -1 \\ c_2 &= -3c_3 + 2 \\ &= 5 \\ c_1 &= -5 + 2 \\ &= -3 \end{aligned}$$

$$[\mathbf{p}(t)]_B = \begin{bmatrix} -3 \\ 5 \\ -1 \end{bmatrix}$$

7. (9 points) Compute the adjugate of the matrix $A = \begin{bmatrix} 0 & -2 & -1 \\ 3 & 0 & 0 \\ -1 & 1 & 1 \end{bmatrix}$.

$$\det A = -3 \begin{vmatrix} -2 & -1 \\ 1 & 1 \end{vmatrix} = 3$$

$$\begin{bmatrix} 0 & -2 & -1 & | & 1 & 0 & 0 \\ 3 & 0 & 0 & | & 0 & 1 & 0 \\ -1 & 1 & 1 & | & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & | & 1/3 & 0 & 0 \\ 0 & -2 & -1 & | & 0 & 0 & 0 \\ 0 & 1 & 1 & | & 0 & 1/3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & | & 0 & 1/3 & 0 \\ 0 & 1 & 1 & | & 0 & 1/3 & 1 \\ 0 & 0 & 1 & | & 1 & 2/3 & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & | & 0 & 1/3 & 0 \\ 0 & 1 & 0 & | & -1 & -1/3 & -1 \\ 0 & 0 & 1 & | & 1 & 2/3 & 2 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{A^{-1}}$

$$\begin{aligned} \text{Adj } A &= \det A \cdot A^{-1} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ -3 & -1 & -3 \\ 3 & 2 & 6 \end{bmatrix} \end{aligned}$$

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8. Consider the matrix $A = \begin{bmatrix} -3 & -2 & 1 & -4 \\ 1 & 3 & 0 & -3 \\ -3 & 4 & -2 & 8 \\ 3 & -4 & 0 & 4 \end{bmatrix} \sim \begin{bmatrix} -3 & -2 & 1 & -4 \\ 1 & 3 & 0 & -3 \\ -9 & 0 & 0 & 0 \\ 3 & -4 & 0 & 4 \end{bmatrix}$

- a) (9 points) Use any valid combination of cofactor expansion and/or row operations to calculate the determinant $\det A$.

$$\det A = 1 \begin{vmatrix} 1 & 3 & -3 \\ -9 & 0 & 0 \\ 3 & -4 & 4 \end{vmatrix} = 9 \begin{vmatrix} 3 & -3 \\ -4 & 4 \end{vmatrix} = 9(12 - 12) = 0$$

- b) (4 points) Find $\det A^T$ using as few steps as possible.

$$\det A^T = 0$$

9. (9 points) Let $W = \left\{ \begin{bmatrix} s+3t \\ s-t \\ 4t \end{bmatrix} : s, t \in \mathbb{R} \right\}$. Show that W is a subspace of $\mathbb{R}^3$

$$W = \left\{ s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} : s, t \in \mathbb{R} \right\}$$
$$= \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} \right\} \text{ which is}$$

a subspace of $\mathbb{R}^3$

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10. The matrices A and B given below are row equivalent. (You do not need to check this.)

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- a) (5 points) Find a basis for the column space of A.

$$B_{\text{col}} = \left\{ \begin{bmatrix} 1 \\ 3 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 5 \\ 2 \\ 8 \end{bmatrix} \right\}$$

- b) (10 points) Find a basis for the null space of A.

$$B\vec{x} = 0$$

$$x_2, x_4 \text{ free}$$

$$x_5 = 0$$

$$x_3 = x_4$$

$$x_1 = -4x_2 - 2x_4$$

$$\vec{x} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -4 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$B_{\text{nul}} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

11. (7 points) Determine whether the set of vectors $B = \{1-t^2, t-t^2, 1-t+t^2\}$ forms a basis for P_2 .

P_2 is isomorphic to $\mathbb{R}_3$ by $a+bt+ct^2 \mapsto \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

Since these vectors are L.I. and $\text{Span } \mathbb{R}_3$,
B is a L.I. set that spans P_2 is a basis for P_2